CS302 90+ FINAL TERM MCQS SPRING 2025

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Question 1

An 8-bit serial-in/parallel-out shift register with value "8" requires ______ clock signals to shift out completely.

Options:

- 1
- 2
- 4
- **8**

Question 2

In a sequential circuit, the next state is determined by:

Options:

- State variable, current state
- Current state, flip-flop output
- **Current state and external input**
- Input and clock signal

Question 3

Divide-by-60 counter in digital clocks uses:

Options:

- **Mod-6, Mod-10**
- Mod-50, Mod-10
- Mod-10, Mod-50
- Mod-50, Mod-6

Question 4

In a NOR-based S-R latch, if both S and R are 0:

Options:

- **Previous state is maintained**
- Output toggles

Question 5

Minimum time input must be stable before clock edge:
Options:

- Set-up time
- **Hold time**
- Pulse Interval time
- Pulse Stability time

Question 6

74HC163 enable pins are:
Options:

- **ENP, ENT**
- ENI, ENC
- ENP, ENC
- ENT, ENI

Question 7

Multiple internal changes due to one input change:
Options:

- Clock Skew
- **Race condition**
- Hold delay
- Hold and Wait

Question 8

________ input overrides ________ input:
Options:

- **Asynchronous, synchronous**
- Synchronous, asynchronous
- PRE, CLR
- CLR, PRE

Question 9

A decade counter is:
Options:

- Mod-3

- Mod-5
- Mod-8
- **Mod-10**

Question 10

Idle asynchronous transmission line is:
Options:

- Logic low
- **Logic high**
- Previous state
- State unused

Question 11

A nibble has ______ bits:
Options:

- 2
- **4**
- 8
- 16

Question 12

Excess-8 code for "-8":
Options:

- 1110
- 1100
- 1000
- **0000**

Question 13

Inverting amplifier gain formula:
Options:

- **Vout/Vin = -Rf/Ri**
- Vout/Rf = -Vin/Ri
- Rf/Vin = -Ri/Vout
- Rf/Vin = Ri/Vout

Question 14

LUT stands for:

Options:

- **Look Up Table**
- Local User Terminal
- Least Upper Time
- None

Question 15

Fundamental gates:

Options:

- AND, NAND, XOR
- OR, AND, NAND
- NOT, NOR, XOR
- **NOT, OR, AND**

Question 16

Total memory supported depends on:

Options:

- Memory organization
- Memory structure
- Decoding unit size
- **Address bus size**

Question 17

Stack is:

Options:

- FIFO
- **LIFO**
- Flash Memory
- Bust Flash

Question 18

Octal "36" + "71" = ______:
Options:

- 213
- **127**
- 123
- 345

Question 19

Example of synchronous input:
Options:

- **J-K input**
- EN input
- PRE
- CLR

Question 20

Clock signal arriving at different times:
Options:

- Race condition
- **Clock Skew**
- Ripple Effect
- None

Question 21

Up/down counter (X=1: count down), current state "1100":
Options:

- 0000
- 1101
- **1011**
- 1111

Question 22

State transition depends on:
Options:

- **Current state and inputs**

- Current state and outputs
- Previous state and inputs
- Previous state and outputs

Question 23

Simplifies next-state logic:
Options:

- State diagram
- Next state table
- State reduction
- **State assignment**

Question 24

J-K flip-flop vs. S-R:
Options:

- **No invalid state**
- Clears when J=K=0
- No pulse transition
- No async inputs

Question 25

Multiplexer with register converts:
Options:

- Serial to parallel
- **Parallel to serial**
- Serial to serial
- Parallel to parallel

Question 26

GAL is a:
Options:

- Non-reprogrammable PAL
- Manufacturer-programmed PAL
- Large PAL
- **Reprogrammable PAL**

Question 27

All columns in a row read/written together:
Options:

- Sequential Access
- MOS Access
- **FAST Mode Page Access**
- None

Question 28

Synchronizing devices with different data rates:
Options:

- ROM
- **FIFO Memory**
- Flash Memory
- Fast Page Access

Question 29

Positive edge-triggered flip-flop changes state on:
Options:

- **Low-to-high clock**
- High-to-low clock
- EN set
- PRE set

Question 30

Frequency counter measures:
Options:

- Pulse width
- **Clock pulses per second**
- Clock pulse range
- None

Question No: 31

Excess-8 code assigns ______ to "+7".

- ( ) 1001
- ( ) 1000
- ( ) 1111
- **(✓) 0000** (Page 34)

Question No: 32

NOR gate is formed by connecting ______.

- ( ) NOT Gate and then OR Gate
- ( ) AND Gate and then OR Gate
- ( ) OR Gate and then AND Gate
- **(✓) OR Gate and then NOT Gate** (Page 47)

Question No: 33

A full-adder has Cin = 0. What are the sum (Σ) and carry (Cout) when A = 1 and B = 1?

- ( ) Σ = 0, Cout = 0
- **(✓) Σ = 0, Cout = 1** (Page 135)
- ( ) Σ = 1, Cout = 0
- ( ) Σ = 1, Cout = 1

Question No: 34

A particular half-adder has ______.

- ( ) 2 INPUTS AND 1 OUTPUT
- **(✓) 2 INPUTS AND 2 OUTPUT** (Page 134)
- ( ) 3 INPUTS AND 1 OUTPUT
- ( ) 3 INPUTS AND 2 OUTPUT

Question No: 35

The four outputs of two 4-input multiplexers (combined to form a 16-input MUX) are connected through a 4-input ______ gate.

- ( ) AND
- **(✓) OR** (Page 171)
- ( ) NAND
- ( ) XOR

Question No: 36

A Field-Programmable Logic Array can be programmed by the user (not the manufacturer).

- **(✓) TRUE** (Page 182)
- ( ) FALSE

Question No: 37

Flip-flops are also called ______.

- ( ) Bi-stable dualvibrators
- ( ) Bi-stable transformer
- **(✓) Bi-stable multivibrators** (Page 228)
- ( ) Bi-stable singlevibrators

Question No: 38

A positive edge-triggered flip-flop changes state on ______.

- **(✓) Low-to-high clock transition** (Page 228)
- ( ) High-to-low clock transition
- ( ) Enable (EN) input set
- ( ) Preset (PRE) input set

Question No: 39

______ is an example of synchronous inputs.

- **(✓) J-K input** (Page 235)
- ( ) EN input
- ( ) Preset (PRE) input
- ( ) Clear (CLR) input

Question No: 40

Glitches due to race conditions can be avoided using ______.

- ( ) Gated flip-flops
- ( ) Pulse-triggered flip-flops
- ( ) Positive-edge triggered flip-flops
- **(✓) Negative-edge triggered flip-flops** (Page 267)

Question No: 41

Synchronous counter design starts from ______.

- ( ) Truth table
- ( ) K-map
- ( ) State table
- **(✓) State diagram** (Page 319)

Question No: 42

A digital clock's hours counter is implemented using ______.

- ( ) Single MOD-12 counter
- ( ) MOD-10 and MOD-6 counters
- ( ) MOD-10 and MOD-2 counters
- **(✓) Single decade counter + flip-flop** (Page 299)

Question No: 43

Given an up/down counter's state diagram, we can determine ______.

- **(✓) Next state from current state** (Page 371)
- ( ) Previous state from current state
- ( ) Both next/previous states
- ( ) Only inputs/outputs

Question No: 44

In ______ machines, outputs depend only on the current state.

- ( ) Mealy
- **(✓) Moore** (Page 332)
- ( ) State Reduction table
- ( ) State Assignment table

Question No: 45

A synchronous decade counter requires ______ flip-flops.

- ( ) 3
- **(✓) 4** (Page 281)
- ( ) 7
- ( ) 10

Question No: 46

A multiplexer + register circuit converts ______.

- ( ) Serial to parallel
- **(✓) Parallel to serial** (Page 356)
- ( ) Serial to serial
- ( ) Parallel to parallel

Question No: 47

Alternative to a MUX + register circuit is ______.

- **(✓) Parallel-in/Serial-out shift register** (Page 356)
- ( ) Serial-in/Parallel-out shift register
- ( ) Parallel-in/Parallel-out shift register
- ( ) Serial-in/Serial-out shift register

Question No: 48

A 4-bit left shift register initially storing "1" will hold ______ after 3 clock pulses.

- ( ) 2
- ( ) 4
- ( ) 6
- **(✓) 8** (Not confirmed)

Question No: 49

An 8-bit serial-in/parallel-out shift register requires ______ clocks to shift out value "8".

- ( ) 1
- ( ) 2
- ( ) 4
- **(✓) 8** (Page 356)

Question No: 50

A 5-bit Johnson counter sequences through ______ states.

- ( ) 7
- **(✓) 10** (Page 354)
- ( ) 32
- ( ) 25

Question No: 51

In ______ counters, the last flip-flop's Q output connects to the first flip-flop's data input.

- ( ) Moore machine
- ( ) Mealy machine
- ( ) Johnson counter
- **(✓) Ring counter** (Page 355)

Question No: 52

DRAM stands for ______.

- **(✓) Dynamic RAM** (Page 407)
- ( ) Data RAM
- ( ) Demodulator RAM
- ( ) None

Question No: 53

If FIFO memory is full, ______.

- ( ) It locks (no new data)
- ( ) New data overwrites old
- ( ) Old data is swapped out
- **(✓) None of the above**

Question No: 54

LUT stands for ______.

- **(✓) Look Up Table** (Page 439)
- ( ) Local User Terminal
- ( ) Least Upper Time Period
- ( ) None

Question No: 55

D/A converter ______ is determined by comparing actual vs. expected output.

- ( ) Resolution
- **(✓) Accuracy** (Page 460)
- ( ) Quantization

- ( ) Missing code

Question No: 56

In a 3-bit synchronous counter, the red rectangle (missing diagram) would be replaced by a ______ gate.

- ( ) AND
- ( ) OR
- ( ) NAND
- ( ) XNOR

Question No: 57

When J-K flip-flop inputs are both 0, ______.

- ( ) It triggers
- ( ) Q=0, Q'=1
- ( ) Q=1, Q'=0
- **(✓) Output remains unchanged** (Page 223)

Question No: 58

A frequency counter ______.

- ( ) Counts pulse width
- **(✓) Counts clock pulses in 1 second** (Page 301)
- ( ) Counts high/low clock ranges
- ( ) None

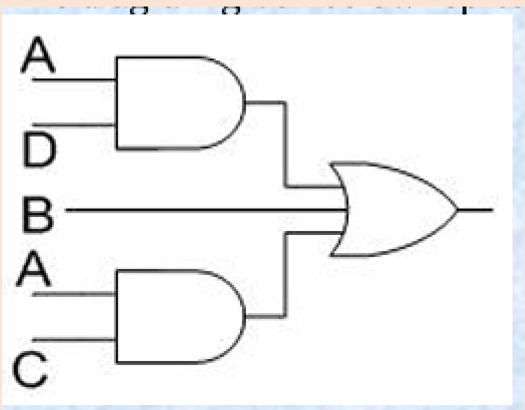
Question No: 59

Stack is short for ______.

- ( ) FIFO memory
- **(✓) LIFO memory** (Page 429)
- ( ) Flash Memory
- ( ) Bust Flash Memory

Question no: 60

The diagram given below represents __________



► Demorgans law

► Associative law

► Product of sum form

► **Sum of product form**

Question No: 61

An 8-bit serial-in/parallel-out shift register contains the value "8". How many clock signals are needed to shift the value completely out?

- 1
- 2
- 4
- **8 (Page 356)**

Question No: 62

The voltage gain of an Inverting Amplifier is given by:

- **Vout / Vin = -Rf / Ri (Page 446)**
- Vout / Rf = -Vin / Ri
- Rf / Vin = -Ri / Vout
- Rf / Vin = Ri / Vout

Question No: 63

In a sequential circuit, the next state is determined by ______ and ______.

- State variable, current state
- Current state, flip-flop output
- **Current state and external input (Page 318)**
- Input and clock signal applied

Question No: 64

The divide-by-60 counter in a digital clock uses two cascading counters:

- **Mod-6, Mod-10 (Page 299)**
- Mod-50, Mod-10
- Mod-10, Mod-50
- Mod-50, Mod-6

Question No: 65

In a NOR gate-based S-R latch, if both S and R inputs are 0, the output:

- **Remains unchanged (Page 221)**
- Resets to 0
- Sets to 1
- Becomes invalid

Question No: 66

A nibble consists of ______ bits.

- 2
- **4 (Page 394)**
- 8
- 16

Question No: 67

Excess-8 code assigns ______ to "-8".

- 1110
- 1100
- 1000
- **0000 (Page 34)**

Question No: 68

LUT stands for ______.

- **Look Up Table (Page 439)**
- Local User Terminal
- Least Upper Time Period
- None of the above

Question No: 69

The three fundamental logic gates are ______.

- AND, NAND, XOR
- OR, AND, NAND
- NOT, NOR, XOR
- **NOT, OR, AND (Page 40)**

Question No: 70

Total memory supported by a digital system depends on ______.

- Memory organization
- Memory structure
- Decoding unit size
- **Address bus size of the microprocessor (Page 430)**

Question No: 71

Stack is short for ______.

- FIFO memory
- **LIFO memory (Page 429)**
- Flash Memory
- Bust Flash Memory

Question No: 72

Adding octal numbers "36" and "71" results in ______.

- 213
- 123
- **127**
- 345

Question No: 73

An example of synchronous inputs is ______.

- **J-K input (Page 235)**
- EN input
- Preset (PRE) input
- Clear (CLR) input

Question No: 74

_______ occurs when clock signals arrive at different times due to propagation delay.

- Race condition
- **Clock Skew (Page 226)**
- Ripple Effect
- None

Question No: 75

In a state diagram, transitions are determined by _______.

- **Current state and inputs (Page 332)**
- Current state and outputs
- Previous state and inputs
- Previous state and outputs

Question No: 76

_______ simplifies the next-state circuit design.

- State diagram
- Next-state table
- State reduction
- **State assignment (Page 335)**

Question No: 77

A 4-bit serial-in/serial-out shift register initially clear stores "1100". After 2 clock pulses (right-most bit first), the output is _______.

- 1100
- 0011
- 0000
- 1111

Question No: 78

The J-K flip-flop differs from the S-R flip-flop because it _______.

- **Has no invalid state (Page 232)**
- Clears when J=K=0

- Ignores pulse transitions
- Rejects asynchronous inputs

Question No: 79

A multiplexer + register circuit converts ______.

- Serial to parallel
- **Parallel to serial (Page 356)**
- Serial to serial
- Parallel to parallel

Question No: 80

A GAL is essentially a ______.

- Non-reprogrammable PAL
- Manufacturer-only programmable PAL
- Large PAL
- **Reprogrammable PAL (Page 183)**

Question No: 81

In ______, all columns in a row are read/written together.

- Sequential Access
- MOS Access
- **FAST Mode Page Access (Page 413)**
- None

Question No: 82

To synchronize devices with different data rates, use ______.

- ROM
- **FIFO Memory (Page 425)**
- Flash Memory
- Fast Page Access Memory

Question No: 83

A flip-flop changes state on ______.

- **Low-to-high clock transition (Page 228)**

- High-to-low transition
- EN input set
- PRE input set

Question No: 84

A frequency counter ______.

- Measures pulse width
- **Counts clock pulses per second (Page 301)**
- Tracks high/low clock ranges
- None

Question No: 85

In sequential circuits, the next state is determined by ______ and ______.

- State variable, current state
- Current state, flip-flop output
- Current state and external input
- **Input and clock signal (Page 305)**

Question No: 86

Flip-flops are also called ______.

- Bi-stable dualvibrators
- Bi-stable transformer
- **Bi-stable multivibrators (Page 228)**
- Bi-stable singlevibrators

Question No: 87

Given an up/down counter's state diagram, we can determine ______.

- **Next state from current state (Page 371)**
- Previous state
- Both next/previous states
- Only inputs/outputs

Question No: 88

A nibble is ______ bits.

- 2
- **4 (Page 394)**
- 8
- 16

Question No: 89

The output of a circuit with +Vcc, input A, and output Y is always ______.

- 1
- 0
- **A**
- Ā

Question No: 90

When both inputs of an edge-triggered J-K flip-flop are set to logic 0:

- The flip-flop is triggered
- Q=0 and Q'=1
- Q=1 and Q'=0
- **Output remains unchanged**
